

Dibakar Saha

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Professional Summary

- Senior Data Scientist with over **6 years** of experience leading, developing, and deploying **AI, ML, and DL** models in **medical imaging** and **healthcare**.
- Expert in **image processing**, **natural language processing**, **signal processing**, and **Generative AI**.
- Adept at leveraging advanced **machine learning** and **deep learning** solutions to drive innovation and efficiency.
- Skilled at providing data-driven insights through proactive analysis and research.
- Passionate about solving **complex challenges** while increasing technical proficiency.
- Committed to delivering impactful projects through a **collaborative** and **forward-thinking** approach.

Technical Skills

- Programming:** Python, C, C++
- ML/DL Frameworks:** PyTorch, TensorFlow, Keras, scikit-learn
- Data Analysis:** pandas, NumPy
- Image Processing:** OpenCV, DICOM, scikit-image, PIL
- NLP:** NLTK, ELMO
- Databases:** FAISS DB, Chroma, MongoDB
- MLOps:** MLflow
- Server & Infrastructure:** Docker, Kubernetes, Linux
- Performance & Scalability:** Model Optimization, GPU Acceleration

Experience

Nference

2018 - 2025

Senior Data Scientist

Text Redaction & DICOM Data Processing | *Python, PyTorch, CNN, YOLO, ResNet50*

- Developed advanced text redaction algorithms for medical images using DICOM data, ensuring compliance with HIPAA standards.
- Implemented high-recall models for text detection in DICOM images, enhancing patient data privacy in clinical AI use cases.

ECG Attribute Prediction & Mortality Estimation | *Custom 1D CNN Models, Python, PyTorch*

- Created 1D CNN models for ECG attribute prediction and mortality estimation, significantly improving clinical risk assessments.

Image Classification & Search Framework | *FAISS, Vector Search, Object Similarity, Object Detection, Visual Analytics*

- Designed frameworks for image classification, search, and object detection, enabling robust visual data analysis in pathology slides.
- Engineered a similarity search system using ResNet50 and FAISS, allowing efficient research on pathology slides.

Statistical Algorithms & DICOM Cohort Prediction | *Data Analysis*

- Developed statistical methods to predict DICOM cohort sizes, boosting large-scale data analysis efficiency.

Image Annotator Tool | *Foundational Models, SegmentAnything Model, Grounding DINO, QWEN-2.5 VL, Python-Flask, WebUI*

- Built a versatile tool supporting PNG, JPG, BMP, DICOM, NIFTI, and MP4, streamlining AI workflows across multiple formats.
- Spearheaded the implementation of SAMv2, boosting video segmentation speed fivefold to accelerate annotation projects.

Natural vs Synthetic Image Classifier | *Deep Learning*

- Developed a classifier to distinguish natural and synthetic images, aiding the detection of AI-generated content.

Opensource Contributions

LLM-based Live Portfolio Chatbot | *Hugging Face Spaces*

- Created an LLM-powered chatbot showcasing a live portfolio, enabling real-time Q&A interactions with visitors as myself.

Sign-Language (769 stars) | *CNN-based Sign Language Recognition*

- Developed a deep learning model that classifies hand gestures in real-time, enabling efficient sign language recognition.

emojify (285 stars) | *Facial Expression Analysis in Python*

- Implemented a tool that maps facial expressions to corresponding emojis, enhancing interactive user experiences.

Simple-OpenCV-Calculator (253 stars) | *Gesture-Controlled Calculator*

- Created a calculator controlled by hand gestures using OpenCV, reducing the need for physical input devices.

SimpleGestureRecognition (130 stars) | *Jupyter Notebook Demonstration*

- Developed a tutorial showcasing gesture recognition techniques in a notebook format for rapid reproducibility.

Hand-Tracking (75 stars) | *Real-Time Deep Learning Project*

- Built a deep learning model for tracking hand positions in real time, enabling more intuitive user interactions.

Virtual-Mouse (51 stars) | *Webcam-Based Gesture Detection*

- Designed a system that transforms hand gestures captured via webcam into mouse commands, facilitating hands-free control.

Education

Bengal College of Engineering and Technology

Bachelor of Computer Science and Engineering

CGPA: 8.7

2014-2018

Durgapur, West Bengal